Gate 1 :(AND) : $D_1 = X \cdot Q_1$

Gate 2: (OR) : $D_2 = Q_1 + Q_2$

Gate 3: (OR) : $Z = X + Q_2$

D FF: next state = D input on next clock edge

Q_1	Q_2	X	$D_1 = X \cdot Q_1$	$D_2 = Q_1 + Q_2$	$Z = X + Q_1$
0	0	0	0	1	0
0	0	1	1	1	1
0	1	0	0	0	1
0	1	1	1	0	1
1	0	0	0	1	0
1	0	1	0	1	1
1	1	0	0	1	1
1	1	1	0	1	1

$D_1 D_2$ = next state name

State	X	$D_1 D_2$ (next)	State Name	Z
A (00)	0	01	B	0
A (00)	1	11	D	1
B (01)	0	00	A	1
B (01)	1	10	C	1
C (10)	0	01	B	0
C (10)	1	01	B	1
D (11)	0	01	B	1
D (11)	1	01	B	1

Transition Diagram

A — $x = 0 / Z = 0 \longrightarrow$ B

A — $x = 1 / Z = 1 \longrightarrow$ D

B — $x = 0 / Z = 1 \longrightarrow$ A

B — $x = 1 / Z = 1 \longrightarrow$ C

C — $x = 0 / Z = 0 \longrightarrow$ B

C — $x = 1 / Z = 1 \longrightarrow$ B

D — $x = 0 / Z = 1 \longrightarrow$ B

D — $x = 1 / Z = 1 \longrightarrow$ B